MILITARY SPECIFICATION

DECODER GROUP OA-3369(*)/APX-7

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope - The equipment covered by this specification shall provide an active readout function of SIF reply-codes within an IFF MARK X SIF radar recognition system.

1.2 Classification - Decoder Group OA-3369(*)/APX-7 shall be of one type and shall consist of the following items.

Items	Type Designation	Max. Allow. Size (inches)	Max. Allow. Wt. (Lbs.)	Appl. Para.
Decoder, Pulse	KY-384(*)/APX-7	15-1/4 x 6-1/2 x 5-3/4	9-1/2	3.4.2
Decoder, Video	KY-364(*)/APX	5-13/16 x 10-1/8 x 8	12	3.4.3
Control-Indicator	C-3707(*)/APX-7	7-1/4 x 7-1/4 x 6	7	3.4.1

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-E-4682	Electron Tubes and Transistors, Choice and Application of
MIL-W-5088	Wiring, Aircraft, Installation of
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for

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MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-T-18303	Test Procedures; Preproduction and Inspection, for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical, Electronic and Associated Equipment
MIL-D-22844	Decoder, Video KY-364(*)/APX

STANDARDS

Military

MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MS 25245	Connections, Input Power Airborne Electronic Equipment; 28-Volt DC, 115-Volt Single Phase AC, or 115/200 Volt 3-Phase AC, or Combinations

2.2 Availability of Documents

(1) When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval

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Supply Depot, Code CDS, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

3. REQUIREMENTS

3.1 Parts and Materials - In the selection of parts and materials fulfillment of major design objectives shall be the prime consideration. In so doing the following shall govern:

(1) Parts and materials requirements shall conform to Specification MIL-E-5400.

(2) Nonrepairable subassemblies, as outlined in Specification MIL-E-5400, shall be used when practicable. The general size of the subassembly, and the mount of circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable. (See paragraph 6.3.)

(3) When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.1.1 Nonstandard Parts and Material Approval - Approval for the use of nonstandard parts and materials shall be obtained as outlined in Specification MIL-E-5400.

3.1.2 Electron Devices - Electron transistors and diodes shall be chosen and applied, and the complements reported as outlined in Specification MIL-E-4682. The complement report must be submitted to the procuring activity for review and approval prior to preproduction testing.

3.2 Design and Construction

3.2.1 General - The equipment shall conform to all applicable requirements of Specification MIL-E-5400 for design construction and workmanship, unless otherwise specified herein.

3.2.2 Total Weight - The total weight of the equipment, excluding cables, shall be a minimum consistent with good design and shall not exceed 28½ pounds.

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3.2.3 Operating Life

3.2.3.1 Operational Stability - The equipment shall operate with optimum performance for 250 hours, continuously or intermittently, without the necessity for readjustment of any controls which are inaccessible during flight.

3.2.3.2 Total Operating Life - The equipment shall have a minimum total operating life of 2000 hours with reasonable servicing and replacement of parts. Parts requiring replacement within this interval and the life of such parts shall be specified by the manufacturer.

3.2.4 Cabling and Connections

3.2.4.1 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with Specification MIL-E-5400.

3.2.4.2 Interconnection Cabling - The equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of Specification MIL-E-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the equipment.

3.2.4.3 Input Power Connection - Input power connections shall be in accordance with MS 25245.

3.2.5 Interchangeability - The equipment shall meet the interchangeability requirements as defined in Specification MIL-E-5400.

3.2.6 Maintenance Provisions and Field Testing - Provisions for maintenance shall be as specified in Specification MIL-E-5400. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance.

3.2.7 Nomenclature and Nameplates - Nomanclature assignment and nameplate approval for equipment identification shall be in accordance with Specification MIL-N-18307.

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3.2.8 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room ambient ($30^{\circ} \pm 10^{\circ}C$)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input power voltage	27.5 ± 0.5 V DC

3.2.9 Service Conditions - The equipment shall operate satisfactorily under the environmental service conditions or reasonable combination of these conditions in accordance with Specification MIL-E-5400 for Class IA equipment, except as modified herein.

3.2.9.1 Vibration - The equipment shall operate satisfactorily when subjected to the vibration requirements of Curves I and II of Specification MIL-E-5400.

3.2.10 Primary Input Power Requirements - The equipment shall meet all applicable requirements of MIL-STD-704 and shall give specified performance from the following power source with characteristics as defined in MIL-STD-704 having limits as specified therein. The power required shall not exceed the specified amounts.

(1) DC Power 28 V, Category "B", 1.5 Amps.

3.2.11 Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of Specification MIL-I-6181.

3.3 Performance - Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme service conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.

3.3.1 Operation - The equipment shall process the information pulses of SIF reply-codes as received in parallel form from the Video Decoder KY-364/APX and provide a direct numerical readout of the translated binary code designation(s). A three channel

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capability shall be provided for the simultaneous reply-code readout of three discrete targets. In addition, the equipment shall have the following capabilities:

a. Accept signals from a modified azimuth-range indicator comprised of range triggers and of three nominal 400 cps signals: (1) a cw signal for phase reference; (2) two amplitude-modulated signals with quadrature modulation; the modulation being a function of the relative instantaneous position of the antenna and PPI cursor;

b. Generate a range-azimuth gate for segregating targets selected for readout, provide controls for adjusting the gated area, and enable the readout of reply-codes only from targets located within the gated area. (Minimum azimuth variation 5 to 12°). (Minimum range variation 12 to 250 usec.)

c. Provide the circuitry to accept the KY-364/APX IFF-video output and to pass this IFF video with the addition of a range-azimuth gate signal to the associated azimuth-range indicator.

3.3.2 Inputs to the Pulse Decoder

3.3.2.1 Read Gate from KY-364/APX - A negative polarity pulse shall be supplied to the equipment, coincident with the application of the information pulses. The pulse shall have the following characteristics:

a.	Duration	0.13 ± 0.05 usec
b.	Amplitude	6 to 8 volts
c.	Baseline	+12 volts
d.	Rise and Decay Times	0.1 usec max

3.3.2.2 Information Pulses from KY-364/APX - The information pulses of SIF reply-codes, when present, shall be furnished in parallel form on twelve wires, and have the following pulse characteristics:

a.	Duration	0.55 ± 0.10 usec
b.	Amplitude	5.5 ± 0.75 volts
c.	Baseline	+12 volts
d.	Polarity	Negative

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e. Rise time 0.4 usec max

f. Decay time 0.3 usec max

3.3.2.3 Azimuth Synchro Voltage from Indicator -

a. Phase Voltage: Frequency: 400 cps (± 20 cps)

	Antenna Pos. Relative to Cursor	
	0° - 180°	90° - 270°
Phase 1	0 to 0.05 v. RMS	26.0 ± 1.5 v RMS
Phase 2	26.0 ± 1.5 v. RMS	0 to 0.05 v RMS

b. Reference Voltage:

26.0 ± 1.5 v (RMS)

3.3.2.4 Range Trigger from Indicator - The range trigger input shall be a pulse in time coincidence with the PPI Range Strobe and shall have the following characteristics:

Amplitude	20 ± 5 volts Normal (10 V Min)
Duration	1.0 ± 0.7 usec
Polarity	Positive
Baseline	Ground
Rise Time	.15 usec max
Decay Time	.25 usec max

The range trigger input shall be available on each video sweep of the associated PPI.

3.3.3 Output from Pulse Decoder - The video output for interconnection to the associated azimuth-range indicator shall be available at a BNC connector. The IFF video as received from the KY-364/APX shall be passed unchanged or shall be passed with the addition of the range-azimuth gate signal generated by the equipment.

3.3.3.1 Range-Azimuth Gate Signal - The gate output shall be a series of pulses, positive polarity at a nominal two volts available mixed with the IFF video.

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3.3.4 Decoding

3.3.4.1 General - When a read gate and a range-azimuth gate are simultaneously present, the equipment shall decode the signals appearing at the twelve parallel information pulse leads from the KY-364/APX and display the SIF code that corresponds to these signals. The requirements for SIF codes shall be as specified in Specification MIL-D-22844.

3.3.4.2 Individual Pulse Shift - If any nominal pulse (0.45 usec duration) of a correct code configuration is positioned ± 0.2 usec or less from nominal position, a proper code readout shall be displayed; if a pulse is shifted ± 0.5 usec or more from nominal position, the code readout shall not reflect this displaced pulse. (The 0.5 usec shifted pulse shall be considered effectively as of zero value.)

3.3.4.3 Decoding at Extreme Temperatures - The decoding tolerances specified in 3.3.4.2 shall apply for equipment operation within the temperature range of 20°C and +40°C. At temperatures beyond this range but within the Class 1A limits of Specification MIL-E-5400, the equipment will operate after a ten-minute warm-up period but with reduced tolerances. The decoding tolerances specified in 3.3.4.2 will change from 0.2 to 0.1 usec and from 0.5 to 0.7 usec.

3.3.4.4 Interleaving - The equipment shall be capable of displaying separately up to four interleaved six-pulse trains or two interleaved twelve-pulse trains.

3.3.5 Control-Indicator Display - The equipment shall provide a three channel readout capability for simultaneous display of target reply-codes. The code designation(s) of three discrete targets shall be displayed as a two digit number(s) when a Mode 1 or Mode 3 or Mode 2-6 pulse reply is received and as a four digit number when a Mode 2-12 pulse reply is received. If only a single target is within the range-azimuth gated area, its code designation shall be displayed on all three channels as a result of three separate interrogations and responses.

3.3.5.1 Code Indicator Devices - The Control-Indicator display shall consist of twelve direct-reading numerical indicating devices positioned so that 3 lines of four digit numbers may be displayed. Each displayed numeral shall be not less than 3/8 inch high. The brightness of the display shall be adjustable by the operator.

3.3.5.2 SIF Code Presentation and Reset - The display of reply-codes occuring within the range-azimuth gated area shall be continuous until automatically reset and cleared for a subsequent readout. Automatic resetting shall be a function of the relative cursor and

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antenna positions and shall occur 320° ± 30° after readout. Between reset and a subsequent readout time, the display shall be blank.

3.3.6 Operation of Equipment - The interconnection and operation of the equipment shall not impair other normal functions of the Video Decoder KY-364/APX.

3.4 Detail Requirements

3.4.1 Control Indicator C-3707/APX-7 - The Control-Indicator shall include the following controls on its front panel:

3.4.1.1 "POWER: ON-OFF" - A two-position toggle switch used to energize (or de-energize) both units of the Decoder Group and when required, provide (or interrupt) relay power for modifying the associated azimuth-range indicator.

3.4.1.2 "DIM-BRT" - This control shall permit adjustment of illumination intensity of the energized code indicator display.

3.4.1.3 "AZIMUTH" (Gate Area) - This control shall permit adjustment of the width of the range-azimuth gated area.

3.4.1.4 "RANGE" (Gate Area) - This control shall permit adjustment of the length on the range-azimuth gated area.

3.4.2 Pulse Decoder KY-384/APX-7 - No operator controls shall be available on the Pulse Decoder. Three screwdriver service adjustments shall be provided. The control labelled "GATE AMPL" shall be used to set the output amplitude of the range-azimuth gate to 2.0 volts. The control labelled "12 V ADJ" shall be used to adjust the internal power supply to correspond with the twelve-volt supply in the KY-364/APX. The control labelled "AZ BAL" shall provide adjustment for the synchro system voltage which may be desirable because of variations in associated equipments.

3.4.3 Video Decoder KY-364/APX - The Video Decoder shall meet the requirements of Specification MIL-D-22844.

3.5 Associated Equipment - The equipment shall operate in conjunction with the following associated items that are not supplied as part of this equipment:

a. Radar Recognition Set AN/APX-7

b. Azimuth-Range Indicator IP-203/APS-20 (Modified)*; or IP-414/APA-125 (Modified)*; or IP-229/APA-56 (Modified)*

* Modifications to Azimuth-Range Indicators are not part of this specification.

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4. QUALITY ASSURANCE PROVISIONS

4.1 General - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

(1) Preproduction Tests

(2) Production Sample Tests

(3) Acceptance Tests

(4) Life Tests

4.2 Preproduction Tests - Preproduction tests shall be made on an equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall consist of the following:

(1) Contractor's Demonstration Tests

(2) Service Approval Tests

4.2.1 Contractor's Demonstration Tests - Contractor's demonstration tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.6. The government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory.

4.2.1.1 The contractor shall submit all data collected in conducting these tests to the procuring agency for review and approval. This data shall include a list of all electrical and electronic parts giving their specified voltage, current and temperature rating and the applied circuit voltage, current and ambient and surface temperatures. The ambient and surface temperature shall be obtained under the extreme high temperature operating condition. This data may be included as part of the preliminary parts list.

4.2.2 Service Approval Tests - At the completion of the contractor's demonstration tests and when requested by the procuring activity, the equipment shall be delivered to a specified government laboratory for additional testing. This additional testing may consist of duplicating tests previously conducted and such other tests as are deemed necessary to determine compliance with all applicable design and performance requirements.

4.2.2.1 Accessory Material - In addition to the complete equipment submitted for service approval tests, the contractor shall also submit the accessory material and design and test data specified in Specification MIL-E-5400. This information shall indicate the physical and electrical characteristics of the equipment and establish the equipment's compliance with applicable requirements.

4.2.3 Scope of Tests - Preproduction tests shall include all tests deemed necessary to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental tests in accordance with Specification MIL-T-5422 and interference tests in accordance with Specification MIL-I-6181.

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4.2.4 Preproduction Approval - Approval of the preproduction model shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction model. Prefabrication of production equipment prior to the approval of the preproduction model is at the contractor's own risk. The approved preproduction model will be returned to the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction model shall not be considered as one of the equipments under the contract; however, it may be reworked by the contractor and submitted for acceptance as a production equipment.

4.2.5 Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests imposed on the preproduction sample. Evidence of non-compliance with the above shall constitute cause for rejection or if already accepted by the government it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Production Sample Tests - When requested by the procuring activity, one equipment shall be selected from the first 10 production equipments submitted for acceptance and sent to a designated government laboratory for tests. The equipment shall be selected by the government inspector after the equipment has successfully passed all individual tests.

4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the approved preproduction sample in design, onstruction, workmanship, performance and quality and that it is capable of meeting all applicable requirements.

4.3.2 Production Sample Approval - Approval of the production sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Upon failure of the production sample to pass any of the tests, delivery of equipment under the contract shall cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.3.3 Reconditioning of Production Test Sample - On completion of the production sample test the equipment shall be reworked by the contractor by replacing all wear or damaged items. After reworking the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests

(2) Sampling Tests

(3) Special Tests

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4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product

(2) Operational Test

(3) Manufacturing Reliability Test

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.4.1.3 Manufacturing Reliability Test - Each equipment shall be operated under the conditions specified herein for a period of 6 hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment.

Temperature	Ambient room
Humidity	Ambient room
Vibration	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ±3 g's

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 6-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to that plane for 5 minutes. Where it is not feasible to vibrate the equipment in 2 directions the vertical direction shall be used. During the 6-hour period of operation following the 10-minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 6-hour period specified above may be composed of two 3-hour periods to conform with standard working hours.

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4.4.2 Sampling Tests - Sampling tests shall be conducted on equipments which have successfully passed the individual tests. Equipments shall be selected for sampling tests by the Government Inspector in accordance with the following:

Quantity of Equipments Offered for Acceptance	Quantity to be Selected For Sampling Test
First 10	1
Next 50	1
Next 75	1
Next 100	1
	1 for each additional 200 or fraction thereof

4.4.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction, production and special tests.

(3) Manufacturing reliability test specified in para. 4.4.1.3 except that the test duration shall be 120 hours with no restriction on the number of failures. However, each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurrence in future equipment.

4.4.3 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling tests.

4.4.3.1 Special Tests Schedule - Selection of equipments for special tests shall be made as follows; except that when a Production Sample is requested and delivered to a government laboratory for test (See para. 4.3) the requirement for a special test on an early production equipment may be omitted.

(1) On an early production equipment.

(2) On an early equipment after an engineering change.

(3) Whenever failure reports or other information indicate additional equipments should be tested. (This will be determined by the procuring activity.)

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4.4.3.2 Scope of Tests - Special tests shall consist of such tests as are approved by the procuring activity. Test procedures previously approved for the preproduction and sampling tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests. As a minimum, the following tests shall be made, except that for those equipments selected as a result of design changes and failure reports, only those tests required to check the characteristics in question need be conducted.

(1) Temperature - Altitude Tests

(2) Vibration Test (60 minutes cycling in each of 3 planes)

(3) Humidity Tests

4.4.4 Equipment Failure - Should a failure occur during either the sampling or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occuring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Life Test - The contractor shall furnish all samples and shall be responsible for accomplishing the life test. The test shall be of 300-hours duration and shall be conducted on equipments that have passed the individual test. The life test shall be performed under the conditions specified in 4.5.1. The life test sample shall be selected by the government inspector in accordance with the following; except that when a production sample equipment is requested and submitted to the procuring activity for test then the first equipment specified below need not be selected or subjected to this test. (Equipments which have successfully passed the sampling tests or special tests may be selected for life tests.)

Quantity of Equipments Offered for Acceptance	Quantity to be Selected For Life Test
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof

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4.5.1 Test Conditions - The life test shall be conducted under the following simulated service conditions.

Temperature	Normal room
Altitude	Normal ground (0 - 5000 ft.)
Humidity	Room ambient
A. C. Voltage	115 ± 5 volts (at lowest applicable frequency)
D. C. Voltage	27.0 ± 2.0 volts

4.5.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.5.3 Performance Check - At approximately 8-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.5.4 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test, including all tube failures. This record shall indicate the following:

(1) Part type number

(2) The circuit reference symbol number

(3) The part function

(4) Name of the manufacturer

(5) Nature of the failure

(6) The number of hours which the part operated prior to failure

4.5.4.1 Failure Report - In the event of a failure, the government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of the test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

4.5.5 Reconditioning of Life Test Samples - An equipment which has been subjected to the life test shall be reconditioned as follows:

(1) On completion of the life test, the equipment shall be reworked by the contractor by replacing all "wear" items. The "wear" items shall be determined by agreement between the contractor and the procuring activity.

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(2) After reworking, the contractor may resubmit the equipment for acceptance.

4.6 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When test procedures are available from previous contracts such procedures may be used when approved by the procuring activity.

4.7 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.8 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the government inspector.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-P-17555.

6. NOTES

6.1 Use -

6.2 Test Values - Normal and limiting values of performance data shall be determined at input voltages 27.5 ± 0.5 V DC and 115 ± 1.0 V AC, as applicable. These data are to be used in testing the equipment at installation points for compliance the minimum acceptable standard of performance.

6.3 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

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5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-E-17555.

6. NOTES

6.1 Test Values - Normal and limiting values of performance data shall be determined at an input voltage of 27.5 ± 0.5 V DC as applicable. These data are to be used in testing the equipment at installation points for compliance with minimum acceptable standard of performance.

6.2 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance of reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.3 As a general rule non-repairable subassemblies should be encapsulated or hermetically-sealed. The number of connections internal to the subassembly should be held to a minimum. Detail parts tolerances and ratings should be so selected that the life of the subassembly is greater than that of a similar repairable one. With few exceptions (such as high voltage power supplies), the non-repairable subassembly should evidence a Mean-time-to-failure greater than 5000 hours, and for many applications this figure must be nearer 50,000 hours.

6.4 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

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(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.5 The parentheses (*), when used in the type designation, shall be replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.2.7. The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

NOTICE - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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